



TECHNICAL DOCUMENTATION

ChatGPT Review Interface

A multi-phase prototype exploring transparent AI reasoning,
source verification, and context-aware responses.

Project — Documentation

Phases — 4 Development Stages

Date — May 2026

OVERVIEW

What We Built

This project is a high-fidelity ChatGPT prototype that goes beyond a standard chat interface. It was built across four progressive development phases, each adding a new layer of features. The core idea is to show users *how* the AI thinks — surfacing its assumptions, inferences, source references, and verification flags — giving users a fully transparent and interactive AI experience.

1

Phase 1 — Foundation & Chat UI

Core interface · Sidebar · Animated text streaming

Built the foundational ChatGPT-style interface from scratch — dark theme, sidebar navigation, user message bubbles, and a streaming "typing" animation that simulates the AI generating its response word by word.

- Collapsible left sidebar with New Chat, search, and recent chat history
- User message bubbles (right-aligned) + AI response area (left-aligned)
- Realistic text streaming animation (words appear one by one, 12ms intervals)
- ChatGPT 4o model selector in the header
- Share and action buttons
- Guest User profile in sidebar footer

2

Phase 2 — AI Transparency Layer

Colour-coded highlights · Source viewer · Inline editing

Added the AI Review system on top of the generated response — a one-click "Review" mode that highlights every claim in the text by category, lets users verify sources, and edit incorrect phrases.

- Three-colour highlight system: Blue (assumed context), Yellow (AI inference), Red (verification issue)
- Right Sidebar Source Viewer — clicking a highlight opens the source document inline
- Edit any flagged phrase via a polished inline modal
- Review Progress Bar tracking how many flags the user has resolved

3

Phase 3 — Reasoning Visualisation

Reasoning chain · Knowledge graph · Auto-correction

Added the Reasoning Chain — a visual graph that shows exactly how the AI connected its sources to produce each inference. Users can see the step-by-step logic and auto-correct flagged claims with a single click.

- Visual reasoning graph — nodes for sources, arrows showing inference logic
- Colour-coded sources: Green (verified), Yellow (uncertain), Red (assumption)
- Tooltip cards on each highlight showing AI inference reasoning
- One-click Auto-correction for red-flagged claims
- Side-by-side comparison view: original AI response vs. user-corrected version

4

Phase 4 — Dynamic Topics & Context

Assumption cards · Guided edits · ChatGPT-style sources

The most advanced phase — the AI now shows its assumptions before generating the answer, lets the user correct them, and dynamically changes the full response based on what the user edits.

- Pre-answer Assumption Cards block — shows 3 assumptions the AI made from the prompt
- Edit button on each assumption card — user can correct AI's interpretation
- Context-aware dual answers — changing an assumption produces a completely different response
- ChatGPT-style inline source badges (grey pills) that expand on hover
- Dynamic footer Sources list that updates based on which answer is shown
- Interactive Goal side-onboarding — guides users to discover the edit feature
- Proceed button replaces the chat input until user confirms assumptions

USER EXPERIENCE

Step-by-Step User Flow

This is the complete journey a user takes through the interface — from landing on the welcome screen, to submitting a prompt, reviewing the AI's assumptions, reading the response, and verifying every claim.

1

User opens the app → Welcome Screen

The app loads with a welcome screen showing the ChatGPT 4o header, the sidebar with "New Chat", and a clean empty chat area. The user is presented as "Guest User" in the sidebar footer.

2

User types a question in the chat input → Sends it

The user types e.g. "Analyze India's EV market and suggest growth opportunities for a startup entering this space in 2026". The chat input shows a send button. When submitted, the user's message appears as a right-aligned bubble.

3

Assumption Cards appear — AI shows its interpretations

Before answering, a box appears in the chat showing 3 assumption cards. These are the interpretations the AI made about the question. The chat input is hidden and replaced with a "Proceed" button. A side tooltip ("Interactive Goal") guides the user to try editing an assumption.

4

User edits an assumption (optional) → Clicks Proceed

The user can click "Edit" on any assumption card, retype it (e.g. changing "domestic passenger market" to "commercial fleet"), and save. If they don't edit, the original interpretation is used. Clicking "Proceed" locks the assumptions and starts the AI response.

5

AI response streams in — with inline source badges

The full AI answer appears word by word (streaming animation). Within the text, grey pill badges (e.g. "BEE India", "Amazon Report") are placed next to specific claims. A "Sources" section appears in the footer actions bar alongside the Copy and Like buttons.

6

User hovers a source badge → Popover expands

Hovering over any grey source pill shows a dark popover card with the source title, domain, and an excerpt. Clicking the badge opens the full Source Document in the right sidebar, just like the Review Mode viewer.

7

User clicks "Review" → Colour-coded highlights activate

Clicking the "Review" button overlays the answer with colour highlights. Blue = assumed context, Yellow = AI inference, Red = verification issue. A Review Progress Bar appears, tracking how many flags are resolved. A colour legend appears below the text.

8

User clicks a highlighted phrase → Tooltip appears

A tooltip card pops up with the flag type, a detailed explanation of why this was flagged, and action buttons. For yellow inferences, a "Show Reasoning Chain" button appears. For red issues, an "Auto-Correct" button appears.

9

User views Reasoning Chain → Visual graph appears

A modal opens with a visual node graph showing exactly how the AI connected its source data (green nodes), uncertain data (yellow nodes), and assumptions (red nodes) to arrive at the highlighted inference. Each connection has a label explaining the logical link.

10

User auto-corrects red flags → Compare view opens

Clicking "Auto-Correct" on a red-flagged phrase replaces it with a verified version. After making edits, the user can click "Compare" to open a side-by-side view of the original AI response vs. the user-corrected version, with differences clearly highlighted.

FEATURE DETAILS

Core Features Explained

A detailed look at each major feature, what it does, and the design decisions behind it.



Assumption Cards

WHAT IT DOES

Before generating any answer, the AI displays 3 cards showing the assumptions it made about the user's question — things like what market is being discussed, what type of company the user represents, or what timeframe is relevant. The chat input is hidden until the user clicks "Proceed", forcing them to engage with the context.

WHY IT MATTERS

In real AI responses, the model silently assumes context that it never tells the user. Surfacing these assumptions lets users correct misinterpretations before the answer is generated — saving them from reading a long irrelevant response and then having to re-prompt.



Editable Assumptions → Context-Aware Dual Answers

WHAT IT DOES

Each assumption card has an Edit button. If the user changes Assumption 1 (e.g. from "Indian domestic passenger market" to "Indian commercial fleet market"), the system detects the edit and swaps out the entire response — different title, different sections, different data, different sources — all pre-prepared to reflect the user's corrected interpretation.

INTERACTIVE GOAL TOOLTIP

A side-onboarding tooltip floats to the right of Assumption 1, guiding users with a specific instruction: "Click Edit on Assumption 1, then change the text to 'You're asking about the Indian commercial fleet market' and save!" This teaches users the feature without any modal or tutorial page.



Three-Colour AI Review System

- Blue — Assumed context (data the AI took for granted)
- Yellow — AI inference (conclusion drawn from imperfect data)
- Red — Verification issue (claim that could not be confirmed)

HOW IT WORKS

Clicking "Review" wraps every flagged phrase in a coloured span element. Clicking a span opens a tooltip showing the category, a detailed explanation, and relevant action buttons. A progress bar tracks how many of the N total flags the user has investigated or resolved.



Reasoning Chain (Visual Knowledge Graph)

WHAT IT DOES

For every yellow (AI Inference) highlight, a "Show Reasoning Chain" button opens a modal containing a visual node graph. Source nodes connect to an Inference node via labelled arrows explaining each logical link. Source quality is shown by node colour: green = verified, yellow = uncertain, red = assumption. Users can immediately see whether the AI's chain of logic is trustworthy.



ChatGPT-Style Source Badges

WHAT IT DOES

Inline grey pill badges (e.g. "BEE India", "MoP Draft") appear directly after specific data claims in the AI's text. Hovering them shows a popover card with source title, domain, and excerpt. Clicking opens the full source in the right sidebar. The footer also shows overlapping circular icons of all unique sources referenced in the answer, with a "Sources" label.

DYNAMIC BEHAVIOUR

The sources shown in the footer are not static. They are calculated in real-time by scanning the rendered HTML for data-source-idx attributes and looking up the active topic's source library. When the user switches to the commercial fleet answer, only the fleet-relevant sources appear.

SUMMARY

Feature Summary Table

A consolidated view of every feature across all four phases — what it does and where to find it in the interface.

Phase	Feature	What It Does
Phase 1	Chat Interface	Dark-themed ChatGPT-style UI with left sidebar, model selector, user bubbles, and AI response area.
Phase 1	Text Streaming	AI response appears word-by-word in real time, simulating live generation.
Phase 1	Sidebar Navigation	Collapsible sidebar with New Chat, search bar, recent chats list, and Guest User profile.
Phase 2	Review Mode	One-click button activates colour-coded highlights over the AI's text by category (assumed / inferred / unverified).
Phase 2	Source Viewer	Clicking any highlighted phrase opens the source document in a right sidebar with full text and key finding excerpts.
Phase 2	Inline Phrase Editor	Any flagged phrase can be manually rewritten by the user via a modal dialog.
Phase 2	Review Progress Bar	Tracks how many of the total AI flags the user has resolved (e.g. 2 / 9 resolved).
Phase 3	Reasoning Chain Graph	Visual node-and-arrow diagram showing how the AI connected sources to produce each inference.
Phase 3	Auto-Correct	One click replaces a red-flagged inaccuracy with the verified, correct version of that claim.
Phase 3	Side-by-Side Compare	Opens a panel showing the original AI response next to the user's corrected version with differences highlighted.
Phase 4	Assumption Cards	3 cards appear before the answer showing AI's interpretations of the question. User must click Proceed to continue.
Phase 4	Editable Assumptions	Each assumption card has an Edit button. Saving a change triggers a completely different AI response.
Phase 4	Context-Aware Dual Answers	Two pre-prepared answers (passenger market vs. commercial fleet) swap in based on the user's assumption edit.
Phase 4	Interactive Goal Tooltip	Side onboarding card guides the user to discover the Edit feature with a specific example instruction.
Phase 4	Inline Source Badges	Grey pills (e.g. "BEE India") appear after claims in the text. Hover for preview, click to open full source.
Phase 4	Dynamic Footer Sources	Overlapping source icon circles appear in the chat footer, dynamically updating to match the active answer's sources.

Phase
4**Follow-Up Questions**

After the main answer, suggested follow-up question chips appear which expand into detailed sub-answers inline.

DESIGN PHILOSOPHY

Every feature in this prototype was built around one core idea: **AI should be transparent, not a black box.** Rather than just showing the user a completed answer, the interface makes the AI's full reasoning process visible and interactive — so users understand what the AI assumed, what it inferred, where it got its data, and where it may have made mistakes. This makes the AI more trustworthy and gives users genuine control over the quality of the output.